GROUP PROJECT

Unknown Group

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INTRODUCTION

Workshop machines need proper monitoring to ensure safe and efficient operation.

Many workshops still record machine information manually, which can lead to mistakes and poor tracking.

This project develops a simple monitoring system using C++ to manage machine information and track usage.

PROBLEM STATEMENT

Problem 1

Difficulty tracking machine information

Problem 2

Electricity usage and cost are not calculated automatically.

Problem 3

Machines may overheat and maintenance can be missed.

PROGRAMMING CONCEPTS USED

Structures

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Arrays

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Functions

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Loops

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SYSTEM FEATURES

- Add new machine information
- Display machine list
- Update machine usage hours
- Check machine service status

ALGORITHM

Start the program.

Display the main system menu.

Ask the user to enter a menu option.

1 ,system will allow the user to register a new equipment record.

2 ,the system will update the daily usage data for a specific machine

3 ,will check and display the safety status of all machines

4 will calculate the total electricity consumption and display the cost report.

if 5,he program will terminate

if valid option,an error message will be displayed.

Exit the program

ALGORITHM

TEST CASE

SEGi College Subang Jaya

COURSE CODE- COURSE NAME
TEST CASE

PROJECT TITLE:	
Test Case ID:	
Tester Name:	
Tester's Background:	
a) Area of specialization:	
b) Position/Job:	
c) Industry:	
d) Age:	

3 expert user	can ask your lecturers to test your system
3 normal user	can be anyone

STEPS	USE CASE	FUNCTION	SCENARIOS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS	REMARKS
1	Add Equipment	addNewItem()	User registers new equipment	ID: EQ04 Type: Grinder Power: 1000 MaxHeat: 85	System successfully adds new equipment	Equipment added successfully	PASS	
2	Update Usage	updateDailyData()	User updates today's machine usage	ID: EQ01 Hours: 4 T	System updates runtime and temperature	Data Updated	PASS	
3	Invalid Input	updateDailyData()	User enters invalid hours	Hours: 30	System shows error message	Error displayed	PASS	
4	Safety Status	checkStatus()	Machine temperature exceeds limit	Temp: 90 MaxHeat:	System shows CRITICAL alert	CRITICAL displayed	PASS	
5	Energy Report	calcElectricity()	System calculates energy usage	Power: 1200W Hour	Correct energy and cost shown	Correct result, --- DAILY ENERGY REPORT --- Total Energy: X kWh Final Total Cost: RM X	PASS	
6	Exit Programt	main()	User selects exit option	Input: 5	Program terminates	Program closed	PASS	

CHALLENGES FACED

Understanding Structures

Learning how to store machine data using structures.

Deciding System Flow

Planning menu steps so the program runs clearly.

Debugging Errors

Finding and fixing coding and syntax mistakes.

CONCLUSION

This project successfully developed a simple workshop equipment monitoring system using C++.
Programming concepts such as structures, arrays, loops, and functions were applied to organize machine data and system operations.
The system helps manage machine information and improves monitoring efficiency in a workshop environment.

THANK YOU

Any questions?